



BAHRIA UNIVERSITY ISLAMABAD

Department of Computer Science

DESIGN & ANALYSIS of ALGORITHMS

ASSIGNMENT # 3

Submitted By

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Program BS Computer Science

Semester ____5th____

Section ____5A____

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Submission Date: ____26/04/2026____

Project Part 1

Select a Puzzle (Medium/Hard Level: #51 till #150) from the following book:

“Algorithmic Puzzles, Anany V. Levitin and Maria Levitin, Oxford University Press”

➤ I selected **Puzzle #81: Celebrity Problem Revisited** from the book.

1. Problem Statement:

A celebrity in a group of n people is a person who knows nobody but is known by everybody else. We can only ask questions of the form:

“Does person A know person B?”

The task is to design an algorithm which identifies the celebrity or decide that no celebrity exists. The book states this problem as Puzzle #81 and asks for the largest number of questions needed for n people.

2. Objective:

The objective is to solve the celebrity problem using two different algorithm design techniques:

1. Exhaustive Search / Brute Force
2. Decrease-and-Conquer

Then both algorithms will be analyzed and compared.

3. Algorithm 1: Exhaustive Search / Brute Force

Idea

Check every person one by one. For each person i , verify:

1. i knows no other person.
2. Every other person knows i .

If both conditions are true, i is the celebrity. Otherwise, continue checking.

Pseudocode:

Algorithm BruteForceCelebrity(Knows, n)

for i ← 1 to n do

 isCelebrity ← true

 for j ← 1 to n do

 if i ≠ j then

 if Knows(i, j) = true then

 isCelebrity ← false

 break

 if isCelebrity = true then

 for j ← 1 to n do

 if i ≠ j then

 if Knows(j, i) = false then

 isCelebrity ← false

 break

 if isCelebrity = true then

 return i

return "No Celebrity"

Time Complexity Analysis:

For each person, the algorithm may check up to $2(n - 1)$ relations.

Total maximum checks:

$$n \times 2(n - 1) = 2n^2 - 2n$$

Worst Case:

Celebrity is last, or no celebrity exists.

Upper bound: $O(n^2)$

Lower bound: $\Omega(n^2)$

Tight Bound: $\Theta(n^2)$

Best Case:

First, we need to check the person is a celebrity, or the person fails that he is a celebrity.

Upper bound: $O(n)$

Lower bound: $\Omega(n)$

Tight Bound: $\Theta(n)$

Space Complexity:

Only a few variables are used in space complexity.

Worst-case space complexity: $\Theta(1)$

Best-case space complexity: $\Theta(1)$

4. Algorithm 2: Decrease-and-Conquer

Idea

Compare two people at a time.

For two people A and B:

- If A knows B, then A cannot be celebrity.
- If A does not know B, then B cannot be celebrity.

So, one person is eliminated in each step. After $n - 1$ questions, only one candidate remains. Then verify that candidate.

Pseudocode:

Algorithm DecreaseConquerCelebrity(Knows, n)

candidate $\leftarrow$ 1

for i $\leftarrow$ 2 to n do

 if Knows(candidate, i) = true then

 candidate $\leftarrow$ i

for j $\leftarrow$ 1 to n do

 if j $\neq$ candidate then

 if Knows(candidate, j) = true then

 return "No Celebrity"

for j $\leftarrow$ 1 to n do

 if j $\neq$ candidate then

 if Knows(j, candidate) = false then

 return "No Celebrity"

return candidate

Time Complexity Analysis:

The selection of candidate will take “n-1” questions

And verification will take up to “2(n-1)” questions

Total:

$$(n - 1) + 2(n - 1) = 3n - 3$$

So, the algorithm is linear.

Worst Case:

Upper Bound: $O(n)$

Lower Bound: $\Omega(n)$

Tight Bound: $\Theta(n)$

Best Case:

In the best case, the algorithm must compare people to reduce the group of one candidate, so that it will need linear work.

Upper Bound: $O(n)$

Lower Bound: $\Omega(n)$

Tight Bound: $\Theta(n)$

Space Complexity:

In space complexity, only one candidate variable and loop variables are used.

Worst-case space complexity: $\Theta(1)$

Best-case space complexity: $\Theta(1)$

5. Comparison Table:

Feature	Brute Force Algorithm	Decrease-and-Conquer Algorithm
Design technique	Exhaustive Search	Decrease-and-Conquer
Main idea	Check every person completely	Eliminate one non-celebrity at each step
Worst-case time	$\Theta(n^2)$	$\Theta(n)$
Best-case time	$\Theta(n)$	$\Theta(n)$
Space complexity	$\Theta(1)$	$\Theta(1)$
Efficiency	Less efficient	More efficient
Suitable for large input	No	Yes
Number of questions	Up to $2n^2 - 2n$	About $3n - 3$

6. Conclusion:

Both the algorithms solve the Celebrity Problem Revisited correctly. The Brute force algorithm is simple but not efficient because it checks every person against every other person, and then in this way, it gives worst case time complexity $\Theta(n^2)$.

The Decrease-and-conquer algorithm is better than Brute force algorithm because it eliminates one impossible candidate after each question and then verifies the final candidate. Its worst-case time complexity is only $\Theta(n)$. Therefore, the decrease-and-conquer algorithm is a better solution.

7. Sources:

- Assignment #3 PDF, DAA BSCS-5A, Spring 2026.
- Levitin, Anany V., and Maria Levitin. *Algorithmic Puzzles*. Oxford University Press, 2011. Used for Puzzle #81: Celebrity Problem Revisited.
- This report was prepared with the help of CHATGPT.